Supplementary Information (ESM 1) for "TrustFedGNN: Auditing and Mitigating Report-Dependent Trust in Federated Graph Learning"

Ngoc-Son-An Nguyen[1], Quang-Vinh Dang[2*], Tuong-Nguyen Huynh[1]

[1]Industrial University of Ho Chi Minh City, Ho Chi Minh City, Vietnam.
[2*]British University Vietnam, Hung Yen, Vietnam.

S1 Campaigns, Environments and Comparator Tuning

Each study in the main text corresponds to a campaign whose runner, per-seed records, status file and acceptance checks are in the code repository (https://github.com/ngngsonan/TrustFedGNN): C20r (fixed-update audit), C20b/c (end-to-end audit), C21 (report-privacy diagnostic), C22 (GuardFed-style tuning), C23, C23b and C26 (controlled fairness poisoning, logged replay and $\alpha = 0$ run), C24 (adaptive access), C25 (Pokec-z known-label control) and C05 (the supporting sweep of Section S4).

All studies run on CPU except the Pokec-z control, which uses one GPU. Within each study every rule shares the same software environment, seeds, partitions and holdout; the repository records the configuration and environment of every run.

GuardFed-style tuning

C22 searched the trust-score temperature τ and selection threshold Γ on tuning seeds 1000–1004 and froze the selected values before confirmation: $(\tau, \Gamma) = (1, 0.03)$ for German fairness poisoning, $(1, 0)$ for Bail fairness poisoning, and $(10, 0.001)$ for a Bail stealth design. Tuning used $K = 5$, one adversary and 20 rounds. The adaptive-access study (C24, $K = 10$, two adversaries, 40 rounds) reuses the two fairness-poisoning selections for all three attack schedules; the stealth selection is not used there. Because

the implementation substitutes the graph holdout for synthetic validation data, the comparator is called GuardFed-style rather than a reproduction of GuardFed.

S2 Report-Channel Privacy Analysis

Each client k releases its two group means with Gaussian noise, $\tilde{\mu}_{k,a} = \mu_{k,a} + \xi_{k,a}$ with $\xi_{k,a} \sim \mathcal{N}(0, \sigma_k^2)$, calibrated by Rényi accounting as described in the main text, and reports $\widetilde{\mathrm{DPD}}_k = |\tilde{\mu}_{k,0} - \tilde{\mu}_{k,1}|$.

Lemma 1 (Upward bias of privatised disparity) *Let $\tilde{\mu}_{k,a} \sim \mathcal{N}(\mu_{k,a}, \sigma_k^2)$ independently. Under demographic parity ($\mu_{k,0} = \mu_{k,1}$), $\widetilde{\mathrm{DPD}}_k$ follows a half-normal distribution [1] with*

$$\mathbb{E}[\widetilde{\mathrm{DPD}}_k] = \tilde{\sigma}_k \sqrt{\frac{2}{\pi}} > 0, \qquad \tilde{\sigma}_k = \sqrt{2}\,\sigma_k. \tag{S1}$$

For weights $\boldsymbol{w} \in \Delta_K$ that are fixed or determined by earlier rounds (measurable with respect to the filtration $\mathcal{F}_{t-1}$),

$$\mathbb{E}\left[\sum_{k=1}^{K} w_k \widetilde{\mathrm{DPD}}_k \,\middle|\, \mathcal{F}_{t-1}\right] = \sqrt{\frac{2}{\pi}} \sum_{k=1}^{K} w_k \tilde{\sigma}_k > 0, \tag{S2}$$

and for weights chosen after seeing the current reports, $\sum_k w_k \widetilde{\mathrm{DPD}}_k \geq \min_k \widetilde{\mathrm{DPD}}_k$, whose expectation is also positive. The noise therefore accumulates across clients instead of cancelling.

Proof $Z = \tilde{\mu}_{k,0} - \tilde{\mu}_{k,1} \sim \mathcal{N}(0, 2\sigma_k^2)$, so $|Z|$ is half-normal with scale $\tilde{\sigma}_k$ and mean $\tilde{\sigma}_k\sqrt{2/\pi}$. Taking the conditional expectation with $\mathcal{F}_{t-1}$-measurable weights gives Eq. (S2). For arbitrary weights on the simplex, the weighted sum of non-negative reports is at least their minimum, which has positive mean. □

The bias comes from the absolute value. A client could instead release the signed difference $\tilde{\mu}_{k,0} - \tilde{\mu}_{k,1}$, which is unbiased. The next result concerns a narrow, single-round Gaussian testing problem and is not a general impossibility result for private fairness mechanisms.

Proposition 1 (Per-round observability bound under local differential privacy) *Let the clients release privatised disparity reports $\widetilde{\boldsymbol{D}} = (\widetilde{\mathrm{DPD}}_1, \ldots, \widetilde{\mathrm{DPD}}_K)$ generated by this release with effective noise standard deviation $\tilde{\sigma}$. For any decision rule $\psi(\widetilde{\boldsymbol{D}}) \in \{0, 1\}$ that uses one round of reports to distinguish H_0: every client's disparity satisfies $\delta_k \leq \tau/2$, from H_1: every $\delta_k \geq 3\tau/2$, the minimax testing error $\mathcal{R}^* = \inf_\psi \max\{\Pr_{H_0}[\psi = 1], \Pr_{H_1}[\psi = 0]\}$ satisfies*

$$\mathcal{R}^* \;\geq\; \frac{1}{2}\left(1 - \frac{\tau\sqrt{K}}{2\tilde{\sigma}}\right). \tag{S3}$$

Table S1 Diagnostic error for the privatized scalar-report channel. Mean residuals under heterogeneous and equal-noise approximations over 200 evaluated rounds; no round was excluded. The residual is measured on a fresh noise draw from each client's post-training model, not on the release BFWA consumed. This diagnostic concerns BFWA's report decision only and is not an end-to-end privacy guarantee.

ϵ	R_{het}	R_{equal}	**Excluded rounds**
0.5	0.4923	0.4936	0.0
1	0.4853	0.4881	0.0
2	0.4724	0.4770	0.0
4	0.4487	0.4568	0.0
8	0.4068	0.4236	0.0
16	0.3361	0.3659	0.0
32	0.2222	0.2789	0.0

Proof Take the two hypotheses $\delta_k^{(0)} = \tau/2$ and $\delta_k^{(1)} = 3\tau/2$ for every client. Le Cam's two-point lemma [2, 3] gives $\inf_\psi \max\{\Pr_{H_0}[\psi = 1], \Pr_{H_1}[\psi = 0]\} \geq \frac{1}{2}(1 - \mathrm{TV}(P_0, P_1))$. Folding is a deterministic map, so by the data processing inequality $\mathrm{TV}(P_0, P_1) \leq \mathrm{TV}(P_0^{\text{Gauss}}, P_1^{\text{Gauss}})$. For K independent Gaussians with mean separation τ and variance $\tilde{\sigma}^2$, $\mathrm{KL}(P_0^{\text{Gauss}} \,\|\, P_1^{\text{Gauss}}) = K\tau^2/(2\tilde{\sigma}^2)$, and Pinsker's inequality $\mathrm{TV} \leq \sqrt{\mathrm{KL}/2}$ gives $\mathrm{TV} \leq \tau\sqrt{K}/(2\tilde{\sigma})$. □

Remark 1 (Scope of the bound) The bound concerns one round, a fixed two-point hypothesis, independent Gaussian releases, and a common effective noise level. Pooling repeated rounds or modeling heterogeneous client noise can materially change it. Table S1 therefore reports the heterogeneous calculation as a diagnostic approximation. The model-update channel is outside this analysis.

These values quantify the specified single-round diagnostic approximation. They do not establish an intrinsic error rate for every private fairness mechanism, and they do not cover the privacy of client model updates.

Table S2 Adaptive-access results averaged over all 40 rounds. S1 is sustained poisoning from round 10, S2 is intermittent poisoning (five rounds on, five off from round 10), and S3 is reference evasion. Unlike the boundary table of the main text, w_{adv} and I_{adv} here average over every round, including those in which the adversaries behave honestly. Means over 10 paired seeds.

Dataset	Access	Rule	w_{adv}	I_{adv}	AUC	DPD
German	S1	FedAvg	0.2020	1.159	0.6460	0.1464
German	S1	root-FLTrust	0.1977	1.128	0.6520	0.1555
German	S1	FU-Alignment	0.1449	0.857	0.6407	0.0802
German	S1	GuardFed-style	0.0410	0.248	0.6608	0.0966
German	S2	FedAvg	0.2020	1.159	0.6601	0.0660
German	S2	root-FLTrust	0.2008	1.143	0.6642	0.0461
German	S2	FU-Alignment	0.2221	1.301	0.6504	0.0584
German	S2	GuardFed-style	0.0583	0.358	0.6529	0.0661
German	S3	FedAvg	0.2020	3.221	0.5904	0.2195
German	S3	root-FLTrust	0.1698	1.058	0.6237	0.1668
German	S3	FU-Alignment	0.0379	0.288	0.6409	0.0605
German	S3	GuardFed-style	0.0508	0.656	0.6288	0.0734
German	benign	FedAvg	—	—	0.6614	0.0662
German	benign	root-FLTrust	—	—	0.6684	0.0420
German	benign	FU-Alignment	—	—	0.6660	0.0418
German	benign	GuardFed-style	—	—	0.6455	0.0797
Bail	S1	FedAvg	0.1909	1.549	0.7381	0.0229
Bail	S1	root-FLTrust	0.1740	1.089	0.7175	0.0159
Bail	S1	FU-Alignment	0.1169	0.604	0.7157	0.0083
Bail	S1	GuardFed-style	0.1486	1.681	0.7140	0.0099
Bail	S2	FedAvg	0.1909	1.729	0.7490	0.0146
Bail	S2	root-FLTrust	0.1750	1.106	0.7218	0.0135
Bail	S2	FU-Alignment	0.1433	0.734	0.7190	0.0092
Bail	S2	GuardFed-style	0.1787	1.947	0.7240	0.0120
Bail	S3	FedAvg	0.1909	59.255	0.5897	0.0642
Bail	S3	root-FLTrust	0.1386	0.903	0.6417	0.0447
Bail	S3	FU-Alignment	0.0365	0.255	0.7090	0.0147
Bail	S3	GuardFed-style	0.0517	2.731	0.6653	0.0165
Bail	benign	FedAvg	—	—	0.7494	0.0116
Bail	benign	root-FLTrust	—	—	0.7174	0.0121
Bail	benign	FU-Alignment	—	—	0.7148	0.0083
Bail	benign	GuardFed-style	—	—	0.7201	0.0070

S3 Adaptive-Access Results Averaged over All Rounds

The benign rows give each method's utility without an attack. A lower attacked DPD should be read together with the method's benign utility and the attack's measured influence.

S4 Supporting Stealth-Poisoning Sweep

This sweep uses a different protocol from the main studies (one local epoch, 15 rounds, a distance-evading attacker) and is reported only as supporting evidence that the allocation behaviour of FU-Alignment persists as the share of adversaries grows. Its FLTrust arm uses an earlier gradient reference rather than the root update and is therefore omitted.

Table S3 Supporting stealth-poisoning sweep across corruption ratios. Bail Recidivism, $K = 10$, 15 rounds, one local epoch, report privatisation off, 10 seeds. Each adversary sends a fairness-poisoned update projected inside 0.85 of the median benign distance and reports zero disparity. Cells give the adversaries' mean aggregation weight; FU-Alignment stays below the uniform share at every ratio, whereas Krum concentrates weight on the adversary. The last two rows give the FU-Alignment minus FedAvg paired difference and its Holm-adjusted Wilcoxon p across the four ratios.

Rule	**Corruption ratio f/K**			
Uniform share	0.10	0.20	0.30	0.40
FedAvg	0.125	0.247	0.354	0.472
Krum	0.947	1.000	1.000	1.000
Multi-Krum	0.111	0.250	0.429	0.661
FLAME	0.102	0.206	0.309	0.410
BFWA	0.847	0.864	0.881	0.898
FU-Alignment	0.069	0.137	0.212	0.292
FU − FedAvg	-0.056	-0.110	-0.142	-0.180
Holm p	0.131	0.039	0.018	0.008

S5 Artifact Verification

Every campaign has a status file that records the commit, the command lines and the SHA-256 of each output. The repository verifier checks that each artifact matches its recorded hash, that every manifest inside it carries the same clean commit, and that the design is complete (cells × seeds), and it recomputes the summary statistics from the per-seed records. All tables in the article and in this document are regenerated from those records by a single script.

References

[1] Leone, F.C., Nelson, L.S., Nottingham, R.B.: The folded normal distribution. Technometrics **3**(4), 543–550 (1961) https://doi.org/10.1080/00401706.1961.10489974

[2] Le Cam, L.: Convergence of estimates under dimensionality restrictions. The Annals of Statistics **1**(1), 38–53 (1973)

[3] Tsybakov, A.B.: Introduction to Nonparametric Estimation. Springer, New York, NY (2009)